

Hypergraph-Based Unaligned Multi-View Clustering via Cluster-Aware Feature Extraction

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Abstract

The Unaligned Multi-view Clustering (UMC) aims to mine distinct cluster structures from unaligned multi-view data where sample features are not aligned across different views. Most existing methods extract low-dimensional representations from high-dimensional raw features, preserving key components for alignment and fusion. However, these feature extraction strategies often neglect the sample cluster structure, yielding the extracted representations with ambiguous cluster distributions that lead to subsequent suboptimal alignment and fusion. To address this issue, we propose a hypergraph-based cluster-aware alignment clustering framework (HG-UMC), which provides cluster-aware hyperedges for the hypergraph neural network, mining cluster-aware representations for subsequent effective alignment and fusion. Specifically, we design a Hyperedge-driven Feature Extraction module that constructs multiple groups of learnable hyperedges to capture the intrinsic clustering structure of samples precisely. These cluster-aware hyperedges are incorporated into the feature extraction process of the hypergraph neural network, enabling it to learn cluster-aware representations. In the subsequent alignment phase, we design a Hyperedge-driven Cluster-level Alignment and Fusion module. This module extracts the similarity structure among intra-cluster hyperedges to serve as the cluster-specific topology for cross-view cluster alignment, and incorporates a cluster-wise cross-attention mechanism to achieve cross-view cluster-level fusion. Extensive experiments on various datasets have verified the superiority of HG-UMC over other state-of-the-art methods.

CCS Concepts

• **Computing methodologies** → **Cluster analysis**.

Keywords

Unaligned Multi-view Clustering, Unaligned Multi-view Data, Hypergraph learning

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The source code of this paper has been made publicly available at https://doi.org/10.5281/zenodo.20365698.

1 Introduction

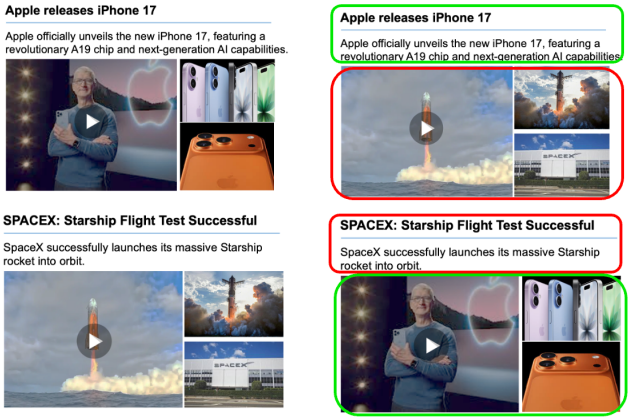
Multi-view data is typically derived from multiple sources or modalities, enabling a more comprehensive data analysis of the objects [5]. For instance, a web page can be simultaneously described by multiple distinct views: textual articles, images, and videos. These views provide complementary information to characterize the same event. Traditional multi-view learning generally assumes that samples across views are strictly one-to-one aligned, as shown in Figure 1 (a). However, in practical scenarios, the collected multi-view data usually shows incorrect correspondences in large-scale and weakly-constrained Internet data, as shown in Figure 1 (b). In such cases, traditional multi-view learning models lose their capability, since they lack the ability to fuse unaligned samples.

The key to learning from unaligned multi-view data lies in how to fuse cross-view information, given the unaligned sample features across different views. Existing unaligned multi-view clustering (UMC) methods usually extract low-dimensional representations from high-dimensional raw features, preserving key components for subsequent alignment and fusion. For instance, Dong et al. [4] extract sample representations through the projection of raw features into a low-dimensional latent subspace, where the extracted representations mitigate the influence of high-dimensional noise and facilitate alignment and fusion based on the geometric consistency of global and local distributions. Ji et al. [11] utilize multiple basis

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(a) Aligned Data (b) Unaligned Data

Figure 1: Illustration of aligned and unaligned multi-view data using web pages as an example. (a) Aligned data, the textual descriptions are consistent with the visual content. (b) Unaligned data, the text descriptions and visual content are mismatched, as highlighted by the colored boxes: the green boxes mark Apple-related content while the red boxes mark SpaceX-related content, showing that they are unaligned.

matrices to project raw features into multi-scale feature spaces, and extract multi-scale feature representations for subsequent cross-view alignment and tensor fusion. However, these methods suffer from a common limitation: their feature extraction strategies usually neglect the sample cluster structure, leading to extracted feature representations with ambiguous cluster distributions, thereby resulting in subsequent suboptimal alignment and fusion.

To address this issue, we propose a hypergraph-based cluster-aware alignment clustering framework (HG-UMC), which provides cluster-aware hyperedges for the hypergraph neural network, mining cluster-aware representations for subsequent effective alignment and fusion. Specifically, we design a Hyperedge-driven Feature Extraction (HFE) module that constructs multiple groups of learnable hyperedges to precisely capture the intrinsic clustering structure of samples. These cluster-aware hyperedges are incorporated into the feature extraction process of the hypergraph neural network (HGNN), enabling it to learn cluster-aware representations. In addition, we design a cluster-aware operator for HGNN, which cluster-wise adjusts the feature extraction process and further enhances the cluster-awareness of the HGNN. In the following alignment and fusion phase, we design a Hyperedge-driven Cluster-level Alignment and Fusion module (HCAF), which extracts the similarity structure among intra-cluster hyperedges to serve as the cluster-specific topology for cross-view cluster alignment. Moreover, in HCAF, we design a cluster-wise cross-attention mechanism to assign higher attention weights to high-confidence samples within aligned clusters, and pull these clusters closer to achieve cluster-level fusion. Finally, the clustering results are obtained by K-means. In summary, the main contributions of this paper lie in:

- We propose a hypergraph-based cluster-aware alignment clustering framework (HG-UMC), which effectively models sample cluster structure and extracts cluster-aware representations for high-quality alignment and fusion.
- We provide **cluster-aware hyperedges** for the hypergraph neural network and design a **cluster-aware operator** to cluster-wise adjust its feature extraction process, mining representations with clear cluster distribution. Based on these representations, we design a **topology-based alignment strategy** and a **cross-attention fusion mechanism**, achieving effective alignment and fusion.
- Extensive experiments on various benchmark datasets have verified the superiority of HG-UMC over other state-of-the-art methods. Furthermore, we conduct ablation studies on our proposed key components, clearly demonstrating their effectiveness within our HG-UMC.

2 Related works

2.1 Multi-view clustering

Multi-view clustering (MVC) aims to extract sample representations from multi-view data and learn the clustering structure on the extracted representations. Existing MVC methods can be broadly categorized into two types: shallow learning methods and neural network-based models. Shallow learning models typically rely on traditional mathematical frameworks, such as Nonnegative Matrix Factorization (NMF) [10, 23, 24, 34, 35, 37, 44] and Graph Learning [7, 12, 22, 25, 28, 29, 39], to uncover linear or near-linear data structures. For instance, Yang et al. [37] generate embedded anchor graphs to model data geometry, and leverage NMF with a correntropy-based loss to extract robust representations that directly determine the cluster partition. Li et al. [18] construct a self-weighted hypergraph to capture high-order correlations, and utilize tensor factorization regularization to derive representations for incomplete data that facilitate the final clustering assignment. Neural network-based models leverage deep architectures to perform nonlinear transformations, enabling representation extraction from complex sample features [2, 21, 32, 36]. For example, Lin et al. [21] employ a cross-view dual prediction mechanism to learn consistent representations from incomplete views, and perform clustering based on the common embeddings learned within this contrastive framework. Wang et al. [32] utilize graph autoencoders to preserve local structural similarity, while incorporating contrastive learning to enforce representation consistency across different views. However, these traditional multi-view clustering models inevitably lose their capability under unaligned scenarios, since they lack the ability to effectively fuse unaligned samples.

2.2 Unaligned multi-view clustering

Unaligned Multi-view Clustering (UMC) aims to mine distinct cluster structures from unaligned multi-view data where sample features are not aligned across different views [4, 9, 11, 20, 30, 33, 38, 43]. The key to learning from unaligned multi-view data lies in how to extract sample representations and fuse cross-view information, given the unaligned sample features across different views. For instance, Yu et al. [38] proposed a Non-negative Matrix Factorization framework that extracts local structure-preserving representations, and establishes cross-view associations by minimizing the structural mismatch between view-specific graphs via a

learnable mapping matrix. Wen et al. [33] focused on structural consistency by constructing individual view-specific graphs to capture intrinsic geometry, and employing a projected fixed-point algorithm to match these topological structures for robust cross-view correspondence. Zhao and Xie [42] extract latent representations via auto-encoders, utilizing a Wasserstein GAN for distribution-level alignment and a global attention mechanism for weighted fusion. Although these methods have achieved competitive performance, they still suffer from the same drawback: their feature extraction strategies often neglect the sample cluster structure, yielding the extracted representations with ambiguous cluster distributions that lead to subsequent suboptimal alignment and fusion.

3 Propose Method

In this paper, we propose a hypergraph-based cluster-aware alignment clustering framework HG-UMC, which provides cluster-aware hyperedges for the hypergraph neural network, mining cluster-aware representations for subsequent cluster-level alignment and fusion. The HG-UMC is decomposed into two stages: **Hyperedge-driven Feature Extraction** and **Hyperedge-driven Cluster-level Alignment and Fusion**. In the first stage, we use autoencoder to extract initial latent sample representations, and construct multiple groups of learnable hyperedges to capture the clustering structure of the initial representations. These cluster-aware hyperedges are incorporated into the feature extraction process of the hypergraph neural network with the designed cluster-aware operator, enabling it to learn cluster-aware representations. In the second stage, we compute the cluster-specific topology for each cluster within the learned cluster-aware representations, and employ the Hungarian algorithm to align all views at the cluster-level based on these topology representations. Based on the alignment result, we incorporate a cluster-wise cross-attention mechanism to achieve effective cross-view cluster-level fusion, ultimately obtaining the fused representation utilized for K-means clustering. Figure 2 illustrates the overview of our proposed method.

3.1 Notations

Given an unaligned multi-view dataset $\mathbf{X} = \{\mathbf{X}^v\}_{v=1}^V$ with V views, where $\mathbf{X}^v = \{\mathbf{x}_i^v\}_{i=1}^N \in \mathbb{R}^{N\times D_v}$ denotes the sample features of v-th view, D_v is the dimension of $\mathbf{X}^v$ and N is the number of samples. The goal of HG-UMC is to fuse multi-view information for separating multi-view data into pre-defined C clusters. Notably, the multi-view sample features cannot be directly fused since $\{(\mathbf{x}_i^p, \mathbf{x}_i^q), p \neq q\}$ are derived from different samples in the unaligned multi-view dataset.

3.2 Hyperedge-driven Feature Extraction

Existing UMC methods' feature extraction strategies often neglect sample cluster structure, where the extracted feature representations usually show loose cluster distribution, causing suboptimal alignment and fusion. In this stage, we combine autoencoders and Hypergraph Neural Networks (HGNN) to extract cluster-aware sample representations for facilitating alignment and fusion. Specifically, the feature extraction process is divided into two steps: (1) Utilizing hyperedges to model sample cluster structure. (2) Incorporating the cluster-aware hyperedges into the HGNN to extract cluster-aware sample representations.

3.2.1 Modeling the cluster structure of samples. In this step, we employ view-specific autoencoders to extract latent sample representations and use hyperedges to model the cluster structure of these representations. Specifically, for the v-th view, the encoding process is denoted by $E^v(\mathbf{X}^v) : \mathbf{X}^v \mapsto \mathbf{Z}_{init}^v \in \mathbb{R}^{N\times d}$, where d is the dimension of the latent space and $\mathbf{Z}_{init}^v = \{\mathbf{z}_i^v\}_{i=1}^N$ denotes the extracted initial latent representations. The decoding process is denoted by $D^v(\mathbf{Z}_{init}^v) : \mathbf{Z}_{init}^v \mapsto \mathbf{X}_{pre}^v \in \mathbb{R}^{N\times D_v}$. To strengthen the capability of autoencoders in extracting high-quality representations, a reconstruction loss is introduced as follows:

$$\mathcal{L}_r = \sum_{v=1}^{V} \|\mathbf{X}^v - D^v(E^v(\mathbf{X}^v))\|_F^2. \tag{1}$$

To explicitly model the cluster structure of representations, for each view v, we first apply K-means to partition the representations into C clusters such that $\mathbf{Z}_{init}^v = \{\mathbf{Z}_{init,1}^v, \ldots, \mathbf{Z}_{init,C}^v\}$, yielding C cluster centroids $\{\mu_c^v\}_{c=1}^C$. Then, for each cluster $\mathbf{Z}_{init,c}^v$, we assign a hyperedge prototype group $\mathbf{P}_c^v = \{\mathbf{p}_{c,m}^v\}_{m=1}^M \in \mathbb{R}^{M\times d}$ consisting of M learnable hyperedge prototypes $p_{c,m}^v$ to capture the key cluster components of $\mathbf{Z}_{init,c}^v$. As shown in Figure 3, compared with the simplified capturing strategy of using a single cluster centroid μ_c^v, our modeling strategy employs a hyperedge prototype group $\mathbf{P}_c^v$, which offers multiple fine-grained hyperedge prototypes $\{\mathbf{p}_{c,m}^v\}_{m=1}^M$, thereby capturing more fine-grained cluster components. Given C learnable hyperedge prototype groups $\{\mathbf{P}_c^v\}_{c=1}^C \in \mathbb{R}^{M\times d}$, we construct the hyperedge prototype matrix $\mathbf{P}^v = \{\mathbf{P}_1^v, \ldots, \mathbf{P}_C^v\} \in \mathbb{R}^{k\times d}$, and compute the soft hyperedge incidence matrix $\mathbf{H}^v = \{\mathbf{h}_i^v\}_{i=1}^N \in \mathbb{R}^{N\times k}$:

$$\mathbf{h}_i^v = \text{softmax}\left(\frac{\cos(\mathbf{z}_i^v, \mathbf{P}^v)}{\tau}\right), \tag{2}$$

where $k = M \times C$, $\cos(\cdot,\cdot)$ denotes the cosine similarity, τ is the temperature, and $\mathbf{H}^v$ can be partitioned into $\mathbf{H}^v = \{\mathbf{H}_1^v, \ldots, \mathbf{H}_C^v\}$ corresponding to the clusters of $\{\mathbf{Z}_{init,1}^v, \ldots, \mathbf{Z}_{init,C}^v\}$. In the above steps, the C hyperedge prototype groups $\{\mathbf{P}_1^v, \ldots, \mathbf{P}_C^v\}$ provide fine-grained cluster components of representations $\mathbf{Z}_{init}^v$, enabling the hyperedge incidence matrix $\mathbf{H}^v$ to encode the affiliation of $\mathbf{Z}_{init}^v$ to C hyperedge groups, i.e., the cluster structure of sample representations to C clusters. Then, $\mathbf{H}^v = \{\mathbf{H}_1^v, \ldots, \mathbf{H}_C^v\}$ serves as a cluster-aware hyperedge incidence matrix and is fed into Hypergraph Neural Networks to extract cluster-aware sample representations.

3.2.2 Extracting cluster-aware sample representations. In this step, we feed $\{\mathbf{Z}_{init}^v\}_{v=1}^V$ and $\{\mathbf{H}^v\}_{v=1}^V$ into the HGNN to extract cluster-aware sample representations $\{\mathbf{Z}^v\}_{v=1}^V$. Specifically, we obtain the extracted representations $\{\mathbf{Z}^v\}_{v=1}^V$ as follows:

$$\begin{aligned}\mathbf{Z}^v &= HGNN(\mathbf{Z}_{init}^v, \mathbf{H}^v) \\ &= \sigma\left((\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{H}^v(\mathbf{D}_e^v)^{-1}(\mathbf{H}^v)^\top(\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{Z}_{init}^v\right),\end{aligned} \tag{3}$$

where $\mathbf{D}_n^v \in \mathbb{R}^{N\times N}$ is the diagonal node degree matrix, $\mathbf{D}_e^v \in \mathbb{R}^{k\times k}$ is the diagonal hyperedge degree matrix, and $\sigma(\cdot)$ denotes the non-linear activation function. Guided by the hyperedge incidence matrix $\{\mathbf{H}^v\}_{v=1}^V$ that encodes the cluster structure, HGNN naturally extracts cluster-aware representations $\{\mathbf{Z}^v\}_{v=1}^V$.

However, in Eq. (3), the cluster-aware ability of the HGNN only depends on the input $\{\mathbf{H}^v\}_{v=1}^V$, and the HGNN by itself does not

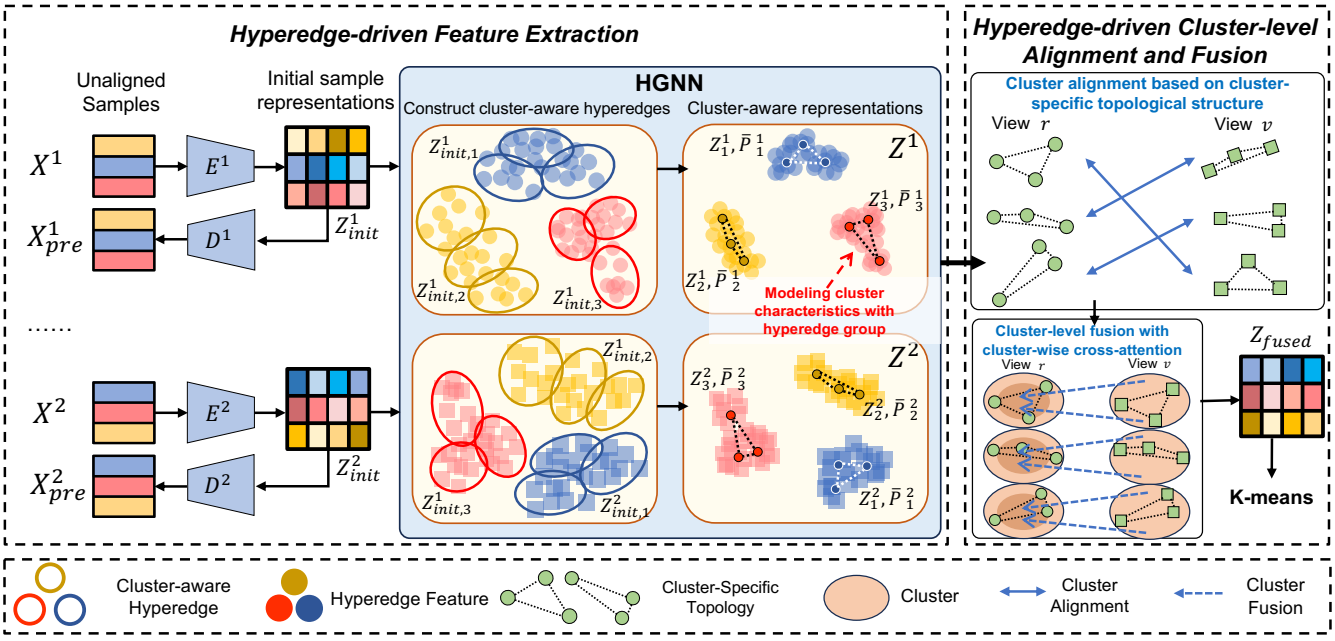


Figure 2: The overview of HG-UMC, which consists of two stages: Hyperedge-driven Feature Extraction and Hyperedge-driven Cluster-level Alignment and Fusion. In the first stage, we extract latent sample representations $\mathbf{Z}^v_{init}$ from unaligned multi-view data $\mathbf{X}^v$ and construct multiple groups of learnable hyperedges to capture the clustering structure of samples. These cluster-aware hyperedges are incorporated into the feature extraction process of the hypergraph neural network, enabling it to learn cluster-aware representations. In the second stage, we leverage cluster-specific topology to align all views with the reference view r at the cluster level, and incorporate a cluster-wise cross-attention mechanism to achieve cross-view cluster-level fusion. Finally, the fused representation $\mathbf{Z}_{fused}$ is used for clustering with K-means.

process the ability. Conventional HGNN typically use a weight operator $\mathbf{\Psi}_w(\cdot)$ to adjust the feature extraction process, treating $(\mathbf{D}_e^v)^{-1}(\mathbf{H}^v)^\top(\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{Z}^v_{init}$ as hyperedge features $\bar{\mathbf{P}}^v$ aggregated from $\mathbf{Z}^v_{init}$ and assign different weights to $\bar{\mathbf{P}}^v$:

$$\mathbf{\Psi}_w(\bar{\mathbf{P}}^v) = \mathbf{W}\bar{\mathbf{P}}^v = \mathbf{W}(\mathbf{D}_e^v)^{-1}(\mathbf{H}^v)^\top(\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{Z}^v_{init}, \tag{4}$$

where $\mathbf{W}$ is a diagonal weight matrix, $\mathbf{\Psi}_w(\bar{\mathbf{P}}^v)$ enables $\bar{\mathbf{P}}^v$ to participate in the feature extraction of HGNN with different weights. Although the weighting mechanism can adjust each hyperedge's importance, it treats each hyperedge independently without the internal correlations within each hyperedge group, which is not enough to extract cluster-aware representations. Therefore, aiming to enhance the cluster-aware capability of the feature extracting process, we cluster-wise partition $\bar{\mathbf{P}}^v$ into C groups $\{\bar{\mathbf{P}}^v_1, \bar{\mathbf{P}}^v_2, \ldots, \bar{\mathbf{P}}^v_C\}$ and substitute the conventional operator with a purposefully designed cluster-aware operator $\mathbf{\Psi}_c(\bar{\mathbf{P}}^v_c)$, which is formulated as:

$$\mathbf{\Psi}_c(\bar{\mathbf{P}}^v_c) = \beta\mathbf{1}_m\bar{\mu}^v_c + (1-\beta)\bar{\mathbf{P}}^v_c, \tag{5}$$

where $\bar{\mu}^v_c = AVG(\bar{\mathbf{P}}^v_c) \in \mathbb{R}^{1\times m}$ is the mean vector of $\bar{\mathbf{P}}^v_c$, $\mathbf{1}_m \in \mathbb{R}^{m\times 1}$ is an all-ones vector for broadcasting, $\beta \in [0,1]$ is the coefficient, and the larger β is, the more similar hyperedge features of $\mathbf{\Psi}_c(\bar{\mathbf{P}}^v_c)$ become. $\mathbf{\Psi}_c(\bar{\mathbf{P}}^v_c)$ enables $\bar{\mathbf{P}}^v = \{\bar{\mathbf{P}}^v_1, \bar{\mathbf{P}}^v_2, \ldots, \bar{\mathbf{P}}^v_C\}$ to participate in the feature extraction with stronger cluster structure and further facilitating the extraction of cluster-aware representations $\mathbf{Z}^v$. The adjusted feature extraction process of HGNN is formulated as:

$$\begin{aligned}\mathbf{Z}^v &= HGNN(\mathbf{Z}^v_{init}, \mathbf{H}^v)\\ &= \sigma\left((\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{H}^v\mathbf{\Psi}_c(\bar{\mathbf{P}}^v)\right),\end{aligned} \tag{6}$$

where $\bar{\mathbf{P}}^v = (\mathbf{D}_e^v)^{-1}(\mathbf{H}^v)^\top(\mathbf{D}_n^v)^{-\frac{1}{2}}\mathbf{Z}^v_{init}$.

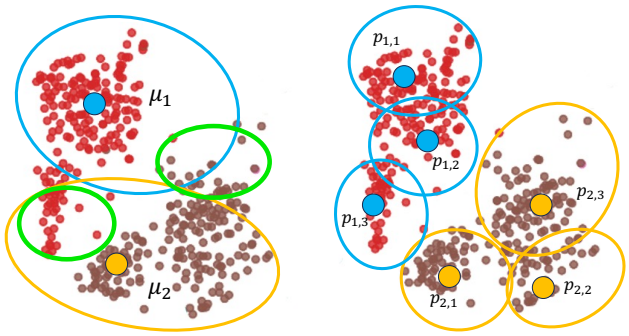
The hyperedge feature $\bar{\mathbf{P}}^v$ is computed from $\mathbf{Z}^v_{init}$ and $\mathbf{H}^v$, where $\mathbf{H}^v = \{\mathbf{H}^v_1, \ldots, \mathbf{H}^v_C\}$ serves as a cluster-aware hyperedge incidence matrix and enables $\bar{\mathbf{P}}^v = \{\bar{\mathbf{P}}^v_1, \ldots, \bar{\mathbf{P}}^v_C\}$ to represent $\mathbf{Z}^v_{init}$'s cluster structure more accurately than initial $\mathbf{P}^v = \{\mathbf{P}^v_1, \ldots, \mathbf{P}^v_C\}$ from *step 3.2.1*. To enhance the capability of $\mathbf{P}^v = \{\mathbf{P}^v_1, \ldots, \mathbf{P}^v_C\}$ in capturing key cluster components, we treat $\bar{\mathbf{P}}^v = \{\bar{\mathbf{P}}^v_1, \ldots, \bar{\mathbf{P}}^v_C\}$ as an optimization target and design the following modeling loss:

$$\mathcal{L}_m = \sum_{v=1}^{V}\sum_{c=1}^{C}\|\mathbf{P}^v_c - \bar{\mathbf{P}}^v_c\|_F^2. \tag{7}$$

The optimized $\mathbf{P}^v$ also refines the hyperedge incidence matrix $\mathbf{H}^v = \{\mathbf{H}^v_1, \ldots, \mathbf{H}^v_C\}$ as Eq. (2), which further enables HGNN to extract cluster-aware representations $\mathbf{Z}^v = \{\mathbf{Z}^v_1, \ldots, \mathbf{Z}^v_C\}$ and update the related hyperedge feature $\bar{\mathbf{P}}^v = \{\bar{\mathbf{P}}^v_1, \ldots, \bar{\mathbf{P}}^v_C\}$.

3.3 Hyperedge-driven Cluster-level Alignment and Fusion

As shown in Figure 2, following the previous stage, we obtain the hyperedge feature groups $\bar{\mathbf{P}}^v = \{\bar{\mathbf{P}}^v_1, \ldots, \bar{\mathbf{P}}^v_C\}$ containing the cluster



(a) Simplified modeling (b) Our modeling

Figure 3: Comparison of cluster modeling strategies. The solid circles represent the cluster centroids in (a) and the hyperedge prototypes in (b), where red and brown colors denote different clusters. The surrounding circles indicate the affiliation of samples to prototypes. (a) The single-centroid approach provides a coarse representation, often failing to capture boundary samples and leading to misclassified samples (highlighted in green). (b) Our multi-hyperedge strategy uses multiple hyperedges to collectively model the cluster, capturing the complex intra-cluster distributions.

structure of clusters $\{\mathbf{Z}_1^v, \ldots, \mathbf{Z}_C^v\}$. Here, the structure of each $\bar{\mathbf{P}}_c^v$ can serve as the topology of $\mathbf{Z}_c^v$ for cluster-level alignment. Since the same-category cluster pair $\{(\mathbf{Z}_c^v, \mathbf{Z}_j^q), v \neq q\}$ shares the consistent topology, **the problem of aligning same-category cluster pairs is transformed into aligning same-topology cluster pairs.** In this stage, we first (1) align the cluster pair $\{(\mathbf{Z}_c^v, \mathbf{Z}_j^q), v \neq q\}$ based on their topology, and then (2) introduce a cluster-wise cross-attention mechanism to fuse each aligned cluster pair $(\mathbf{Z}_c^v, \mathbf{Z}_j^q)$.

3.3.1 Aligning the cluster pair based on their topology. To align the cluster pair $\{(\mathbf{Z}_c^v, \mathbf{Z}_j^q), v \neq q\}$ based on their topology, we first compute the topology representation $\mathbf{t}_c^v$ of each cluster $\mathbf{Z}_c^v$:

$$\mathbf{t}_c^v = \mathrm{eig}(\mathbf{S}_c^v) = \mathrm{eig}\left(\bar{\mathbf{P}}_c^v (\bar{\mathbf{P}}_c^v)^\top\right), \tag{8}$$

where $\mathbf{S}_c^v = \bar{\mathbf{P}}_c^v (\bar{\mathbf{P}}_c^v)^\top \in \mathbb{R}^{M \times M}$ represents the key topology structure of cluster $\mathbf{Z}_c^v$, and $\mathrm{eig}(\mathbf{S}_c^v)$ further extracts M eigenvalues of $\mathbf{S}_c^v$ as the core topology representation $\mathbf{t}_c^v \in \mathbb{R}^{1 \times M}$ of cluster $\mathbf{Z}_c^v$ [27].

Then, after extracting the topology representation $\mathbf{t}_c^v$ of each cluster $\mathbf{Z}_c^v$, we construct a cluster-wise representation matrix $\mathbf{T}^v = \{\mathbf{t}_c^v\}_{c=1}^C \in \mathbb{R}^{C \times M}$ for each view v, and achieve cross-view cluster-wise alignment based on $\{\mathbf{T}^v\}_{v=1}^V$. Specifically, we select a reference view r and align pairs $\{(\mathbf{T}^r, \mathbf{T}^v), 1 \leq v \leq V, v \neq r\}$ for achieving cross-view alignment, where r is set to the view with the largest original feature dimension, as it typically offers a more comprehensive description of samples and a more representative cluster-wise representation $\mathbf{T}^r$ for facilitating cross-view cluster-wise fusion. To align $\{(\mathbf{T}^r, \mathbf{T}^v), v \neq r\}$, we compute the pairwise structural dissimilarity to construct a cost matrix $\mathbf{Q}^{rv} \in \mathbb{R}^{C \times C}$, where $\mathbf{Q}_{ij}^{rv} = \|\mathbf{t}_i^r - \mathbf{t}_j^v\|_2$. Each entry in $\mathbf{Q}^{rv}$ represents an alignment cost, implying that higher values correspond to clusters with dissimilar topological structures. We then minimize the total cost $\mathrm{tr}(\mathbf{\Pi}^{rv}(\mathbf{Q}^{rv})^\top)$ using the Hungarian algorithm [16] to match highly similar clusters, yielding the optimal permutation matrix $\mathbf{\Pi}^{rv}$:

$$\begin{aligned} \mathbf{\Pi}^{rv} = \arg\min_{\mathbf{\Pi}^{rv} \in \{0,1\}^{C \times C}} & \mathrm{tr}(\mathbf{\Pi}^{rv}(\mathbf{Q}^{rv})^\top) \\ \text{s.t.} \quad & \mathbf{\Pi}^{rv}\mathbf{1} = \mathbf{1}, (\mathbf{\Pi}^{rv})^\top \mathbf{1} = \mathbf{1}, \end{aligned} \tag{9}$$

where $\mathbf{\Pi}_{ij}^{rv} = 1$ indicates that the i-th cluster in the reference view r is matched to the j-th cluster in the v-th view. Consequently, we obtain the set of aligned cluster pairs with consistent topology, denoted as $\{(\mathbf{Z}_i^r, \mathbf{Z}_j^v) \mid \mathbf{\Pi}_{ij}^{rv} = 1, v \neq r\}$, where $\mathbf{Z}_i^r \in \mathbb{R}^{N_i^r \times d}$.

3.3.2 Fusing each aligned cluster pair. To effectively fuse each aligned cluster pair in $\{(\mathbf{Z}_i^r, \mathbf{Z}_j^v) \mid \mathbf{\Pi}_{ij}^{rv} = 1, v \neq r\}$, we introduce a cross-attention mechanism that assigns higher weights to high-confidence samples in clusters, thereby achieving more robust and efficient cluster-level fusion. Specifically, for each aligned pair $(\mathbf{Z}_i^r, \mathbf{Z}_j^v)$, we compute the cross-attention weights $\mathbf{A}_{j,i}^v \in \mathbb{R}^{N_j^v \times N_i^r}$ from $\mathbf{Z}_j^v$ to $\mathbf{Z}_i^r$, which is formulated as follows:

$$\mathbf{A}_{j,i}^v = \mathrm{softmax}\left(\frac{\mathbf{Z}_j^v (\mathbf{Z}_i^r)^\top}{\tau}\right), \tag{10}$$

where τ denotes the temperature coefficient that controls the sharpness of the attention distribution, and the softmax operation is applied to normalize the similarity scores $\mathbf{Z}_j^v (\mathbf{Z}_i^r)^\top$ to obtain attention weights $\mathbf{A}_{j,i}^v$. Based on this, for each aligned pair in $\{(\mathbf{Z}_i^r, \mathbf{Z}_j^v) \mid \mathbf{\Pi}_{ij}^{rv} = 1, v \neq r\}$, we pull $\mathbf{Z}_j^v$ closer to $\mathbf{A}_{j,i}^v \mathbf{Z}_i^r$, which enables $\mathbf{Z}_i^r$ to participate in $(\mathbf{Z}_i^r, \mathbf{Z}_j^v)$'s fusion with adaptive attention weights, and then significantly facilitates cross-view cluster-level fusion. Consequently, the fusion loss is formulated as:

$$\mathcal{L}_f = \frac{1}{C} \sum_{\substack{1 \leq v \leq V \\ v \neq r}} \sum_{\mathbf{\Pi}_{ij}^{rv}=1} \left(\|\mathbf{Z}_j^v - \mathbf{A}_{j,i}^v \mathbf{Z}_i^r\|_F^2 + \|\mathbf{t}_j^v - \mathbf{t}_i^r\|_2^2 \right). \tag{11}$$

The first term achieves cluster-level fusion by minimizing the discrepancy for each aligned cluster pair $(\mathbf{Z}_j^v, \mathbf{Z}_i^r)$. The second term enforces topological consistency of $(\mathbf{Z}_j^v, \mathbf{Z}_i^r)$ to further promote cross-view cluster-level fusion.

Finally, we fuse each aligned cluster pair by utilizing the cross-attention weight to synthesize a weighted feature replica in view v. Specifically, similar to Eq. (10), we compute the cross-attention weights $\tilde{\mathbf{A}}_{j,i}^v \in \mathbb{R}^{N_i^r \times N_j^v}$ from $\mathbf{Z}_i^r$ to $\mathbf{Z}_j^v$, which is formulated as:

$$\tilde{\mathbf{A}}_{j,i}^v = \mathrm{softmax}\left(\frac{\mathbf{Z}_i^r (\mathbf{Z}_j^v)^\top}{\tau}\right). \tag{12}$$

For the i-th cluster in view r and its aligned clusters in view v $(v \neq r)$, their fused sample representation $\mathbf{Z}_{fused,i} \in \mathbb{R}^{N_i^r \times d}$ is formulated as:

$$\mathbf{Z}_{fused,i} = \frac{1}{V}\left(\mathbf{Z}_i^r + \sum_{\substack{1 \leq v \leq V \\ v \neq r}} \tilde{\mathbf{A}}_{j,i}^v \mathbf{Z}_j^v\right), \tag{13}$$

where $\tilde{\mathbf{A}}_{j,i}^v \mathbf{Z}_j^v$ acts as a feature replica for view v, reorganized by attention weights to match the reference cluster.

The whole loss function in our method is represented as:

$$\mathcal{L} = \mathcal{L}_r + \lambda \mathcal{L}_m + \gamma \mathcal{L}_f, \tag{14}$$

where λ and γ are trade-off coefficients.

Algorithm 1 The Training Process of HG-UMC.

Input: Unaligned multi-view data $\mathbf{X} = \{\mathbf{X}^v\}_{v=1}^V$, number of clusters C, number of hyperedges M in each group, model training epochs T.
Output: Clustering results
Process:
Initialize autoencoders $\{E^v(\cdot), D^v(\cdot)\}_{v=1}^V$.
Initialize the hyperedge prototypes $\{\mathbf{P}^v\}_{v=1}^V$.
for epoch $t = 1$ to T **do**
  Stage 1 (Hyperedge-driven Feature Extraction):
  for view $v = 1$ to V **do**
    Train autoencoder by Eq.(1) and obtain $\mathbf{Z}_{init}^v$.
    Construct hyperedge incidence matrix $\mathbf{H}^v$ by Eq.(2).
    Extract cluster-aware representation $\mathbf{Z}^v$ by Eq.(6).
  end for
  Stage 2 (Hyperedge-driven Alignment and Fusion):
  Align $\{\mathbf{Z}^v\}_{v=1}^V$ by Eq.(9).
  Fuse aligned $\{\mathbf{Z}^v\}_{v=1}^V$ into $\mathbf{Z}_{fused}$ by Eq.(11).
end for
Perform K-means clustering on $\mathbf{Z}_{fused}$.

3.4 Optimization

Algorithm 1 summarizes the training process of our proposed HG-UMC model. HG-UMC, consisting of multiple autoencoders and hypergraph neural networks, is optimized by a gradient descent algorithm. Specifically, the autoencoders and hypergraph neural networks are trained for sample reconstruction and feature extraction. The autoencoders are optimized by Eq. (1) to extract low-dimensional and informative features. Meanwhile, the hypergraph neural networks are optimized by Eq. (7) to accurately model each cluster by leveraging hyperedges. Subsequently, Eq. (11) is employed to perform cross-view cluster-level fusion, yielding the final fused representations. Finally, the K-means algorithm is applied to these fused representations to obtain the clustering results.

Table 1: Statistics of Benchmark Datasets

Dataset	Samples	Views	Dimensions
BBCSports	544	2	3183/3224
REU	1,200	5	2000/2000/2000/2000/2000
Caltech-5V-7	1,400	5	40/254/1984/512/928
Handwritten	2,000	6	240/76/216/47/64/6
Hdigit	10,000	2	784/256
Cifar10	50,000	3	512/2048/1024

4 Experiments

4.1 Experimental Settings

Datasets: To comprehensively validate our HG-UMC, we conduct experiments on six widely used multi-view benchmark datasets, including **BBCSports** [8], **REU** [1], **Caltech-5V-7** [6], **Handwritten** [31], **Hdigit** [3], and **Cifar10** [40]. The specific structure of these datasets is listed in Table 1.

Compared Methods: To comprehensively verify the superiority of **HG-UMC**, we employ eight state-of-the-art unaligned multi-view clustering methods for comparative experiments, including **DAGF** (2025) [14], **MGCCFF** (2025) [41], **JHCL** (2025) [13], **OpVuC** (2024) [4], **FUMC** (2024) [17], **T-UMC** (2022) [20], **UPMGC-SM** (2023) [33], **MVC-UM** (2021) [38].

Evaluation metrics: We employ four standard metrics to evaluate the clustering performance, including Accuracy (ACC), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and Purity (PUR), whose detailed definitions are illustrated in [19]. For all these metrics, the higher values indicate better performance.

Implementation details: HG-UMC trains 100 epochs by using Adam optimizer [15] with a learning rate of 0.003 in PyTorch [26] deep learning framework. The encoder E^v and decoder D^v are respectively formulated by Multi-Layer Perceptrons with dimensions $\{D_v, 500, 500, 2000, 512, d\}$ and $\{d, 2000, 500, 500, D_v\}$, where the activation function is ReLU and d is set to 32. The hyperparameters β and τ are set to 0.8 and 0.1, respectively. All experiments are implemented on the same machine with the Intel(R) Xeon(R) Gold 6148 2.40GHz CPU, 8 GeForce RTX 3090 GPUs, and 512GB RAM.

4.2 Experimental results

Table 2 records the experimental comparisons on six datasets, where the **best** and the second-best performance are highlighted in bold and underlined, respectively. In addition, Figure 4 illustrates the visualization of clustering results of our method on the Handwritten datasets. According to Table 2 and Figure 4, we can observe that:

(1) In Table 2, our HG-UMC is superior to all comparing methods on all evaluation metrics, and has a significant leading gap compared with the second-best methods. Especially on the *Caltech-5V-7* dataset, the improvements over the second-best method are 15.43%, 20.26%, 21.44%, and 14.86% on ACC, NMI, ARI, and PUR, respectively. These experimental results strongly demonstrate the effectiveness and superiority of HG-UMC, and we attribute such success to our designed hypergraph-based cluster-aware alignment framework, which provides cluster-aware hyperedges for the hypergraph neural network, mining cluster-aware representations for subsequent effective alignment and fusion.

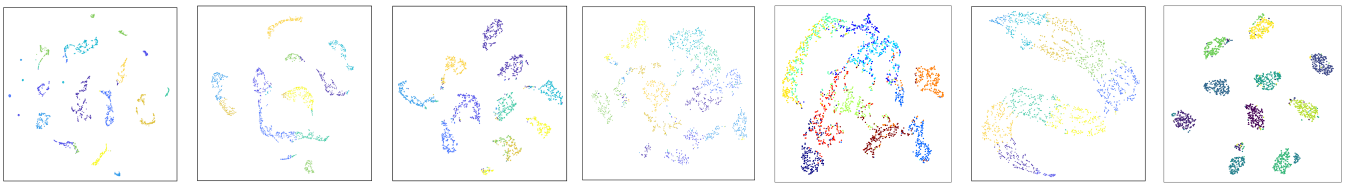
(2) In Figure 4, we select six representative unaligned multi-view clustering methods to conduct the t-SNE visualization comparisons of clustering results with our proposed HG-UMC. According to the visualization result, the compared baseline methods exhibit varying degrees of ambiguous cluster distribution, which directly leads to indistinct semantic representations of the clusters. In contrast, we can observe that our HG-UMC exhibits a clearer and more cohesive cluster structure than other methods, which demonstrates the superiority of HG-UMC in fusing cluster-aware representations from unaligned multi-view data.

4.3 Model analysis

Parameter sensitivity analysis: We experimentally evaluate the effect of hyperparameters γ and λ on the clustering performance of HG-UMC. Figure 5 shows the values of ACC metric on the *Handwritten* and *Hdigit* datasets as γ and λ vary from 10^{-3} to 10^3. It can be observed from the results that HG-UMC is insensitive to λ in

Table 2: Comparative results between HG-UMC and 8 state-of-the-art methods on six datasets. The best results are highlighted in bold, and the second-best are underlined. "-" means that the code can't be run due to its excessive time or space complexity.

Dataset	Metric	MVC-UM	T-UMC	UPMGC-SM	OpVuC	FUMC	DAGF	MGCCFF	JHCL	Ours
BBCSports	ACC	0.4100	0.4543	0.4492	0.4706	0.5978	0.3566	0.3971	0.3647	**0.7087**
	NMI	0.1800	0.1655	0.2948	0.2248	0.3176	0.0154	0.1356	0.0325	**0.3819**
	ARI	0.0698	0.1181	0.1933	0.1783	0.2751	0.0011	0.0729	0.0015	**0.3342**
	PUR	0.4233	0.4688	0.5214	0.5441	0.6120	0.3603	0.4055	0.3721	**0.7087**
REU	ACC	0.4083	0.4425	0.5440	0.3317	0.2327	0.1707	0.3692	0.1698	**0.6250**
	NMI	0.2373	0.2086	0.3149	0.1408	0.0230	0.0089	0.1590	0.0084	**0.3690**
	ARI	0.1756	0.1783	0.2682	0.0870	0.0136	0.0000	0.1073	0.0000	**0.3255**
	PUR	0.4619	0.4492	0.5588	0.4575	0.2405	0.1715	0.3842	0.1710	**0.6250**
Caltech-5V-7	ACC	0.2785	0.4079	0.7293	0.3279	0.2044	0.1709	0.3857	0.1725	**0.8836**
	NMI	0.1038	0.3271	0.5856	0.1229	0.0269	0.0074	0.2543	0.0074	**0.7882**
	ARI	0.0875	0.3180	0.5424	0.0820	0.0137	0.0003	0.2002	0.0003	**0.7568**
	PUR	0.3014	0.4393	0.7350	0.3851	0.2098	0.1744	0.4381	0.1754	**0.8836**
Handwritten	ACC	0.7465	0.7720	0.6404	0.1465	0.1946	0.7835	0.7248	0.6410	**0.8840**
	NMI	0.7230	0.6703	0.6485	0.0181	0.0652	0.8054	0.7543	0.6531	**0.8065**
	ARI	0.6305	0.6564	0.5043	0.0052	0.0431	0.6973	0.5622	0.5051	**0.7714**
	PUR	0.7465	0.7720	0.6652	0.1705	0.2004	0.8250	0.7248	0.6410	**0.8840**
Hdigit	ACC	0.4627	0.4993	0.4487	0.3994	0.3603	0.3577	0.4167	0.2673	**0.6745**
	NMI	0.4418	0.4387	0.4187	0.3151	0.3355	0.4526	0.4609	0.2708	**0.6119**
	ARI	0.3987	0.3708	0.2897	0.2054	0.0216	0.2193	0.0626	0.0772	**0.5191**
	PUR	0.5031	0.5398	0.4887	0.4174	0.4289	0.3585	0.5182	0.2779	**0.6808**
Cifar10	ACC	0.4627	0.6300	-	0.8008	0.2174	-	0.1580	-	**0.8483**
	NMI	0.4418	0.4096	-	0.6872	0.0892	-	0.0382	-	**0.7081**
	ARI	0.3987	0.4049	-	0.6284	0.0747	-	0.0120	-	**0.6992**
	PUR	0.5031	0.6300	-	0.8008	0.2191	-	0.1675	-	**0.8483**



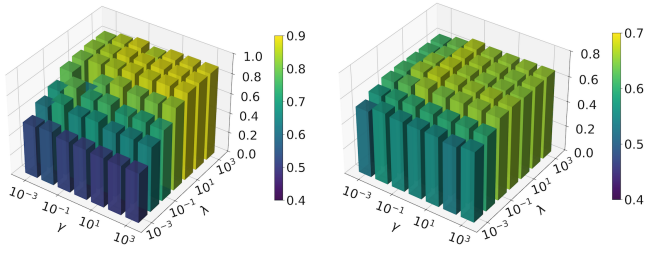
(a) JHCL (b) UPMGC-SM (c) DAGF (d) T-MUC (e) MGCCFF (f) MVC-UM (g) Ours

Figure 4: The t-SNE visualizations of the clustering results comparing different methods on the *Handwritten* dataset.

the range from 10^1 to 10^3 and to γ in the range from 10^{-1} to 10^3. In our experiments, we set $\gamma = 10$ and $\lambda = 300$.

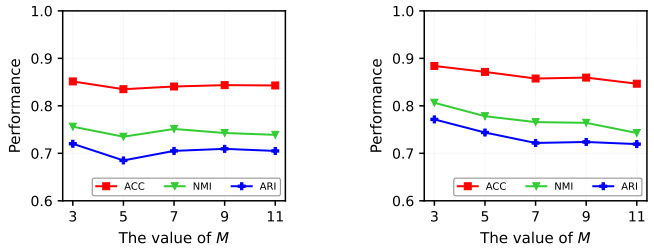
Furthermore, we experimentally evaluate the effect of the hyperedge number M within each hyperedge group on the overall clustering performance of our proposed model HG-UMC. Figure 6 graphically illustrates the variations in ACC, NMI, and ARI metric values on the *Handwritten* and *Caltech-5V-7* datasets. Based on the experimental results, it can be observed that the clustering performance of HG-UMC shows a slight decline as M increases but with minimal fluctuations, indicating that the model is insensitive to the parameter M. In our experiment, M is set to 3.

Convergence analysis: Figure 7 shows the convergence curves of HG-UMC on four datasets: *Handwritten*, *Caltech-5V-7*, *Hdigit*, and *Cifar10*, where the values of loss and evaluation metrics are illustrated in each subfigure. As shown in Figure 7, we can clearly observe that the training loss drops significantly at the beginning of the iteration process and then gradually converges to a stable value as the number of iterations increases. The evaluation metrics exhibit fluctuations during the early stages and gradually stabilize in the late stages. These results effectively verify the convergence of our proposed HG-UMC.



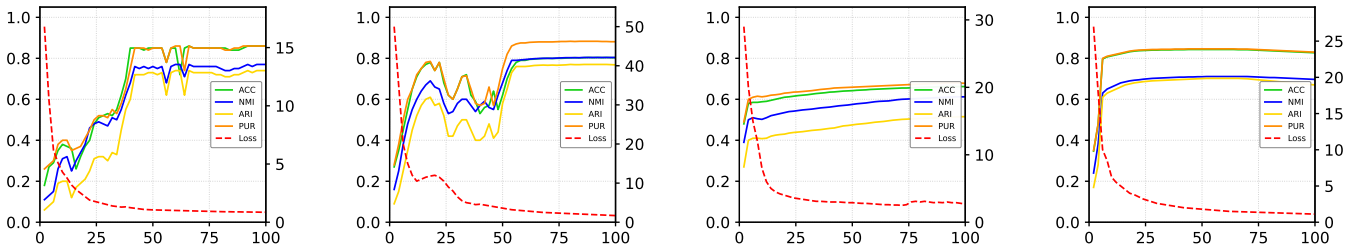
(a) Handwritten (b) Hdigit

Figure 5: Parameter sensitivity analysis of γ and λ on two datasets, where the vertical axis represents the ACC value.



(a) Caltech-5V-7 (b) Handwritten

Figure 6: Parameter sensitivity analysis of M on two datasets.



(a) Handwritten (b) Caltech-5V-7 (c) Hdigit (d) Cifar10

Figure 7: The convergence analysis on *Handwritten*, *Caltech-5V-7*, *Hdigit* and *Cifar10*, datasets.

Table 3: Ablation studies on loss functions of HG-UMC on *Hdigit* and *Handwritten* datasets.

	Loss			Hdigit				Handwritten			
	$\mathcal{L}_r$	$\mathcal{L}_m$	$\mathcal{L}_f$	ACC	NMI	ARI	PUR	ACC	NMI	ARI	PUR
(A)		✓	✓	0.6342	0.5672	0.4775	0.6484	0.8502	0.7683	0.7339	0.8502
(B)	✓		✓	0.5422	0.4253	0.3310	0.5683	0.7895	0.6604	0.5972	0.7895
(C)	✓	✓		0.4542	0.3101	0.2228	0.4762	0.2285	0.1065	0.0525	0.2360
(D)			✓	0.5336	0.4187	0.3238	0.5527	0.7640	0.7009	0.6288	0.7640
(E)	✓	✓	✓	**0.6745**	**0.6119**	**0.5191**	**0.6808**	**0.8840**	**0.8065**	**0.7714**	**0.8840**

Table 4: Ablation studies on model components of HG-UMC on *Hdigit* and *Handwritten* datasets.

	Components			Hdigit				Handwritten			
	C_h	C_t	C_i	ACC	NMI	ARI	PUR	ACC	NMI	ARI	PUR
(a)		✓	✓	0.5898	0.5269	0.4224	0.6267	0.8445	0.7511	0.7029	0.8445
(b)	✓		✓	0.6423	0.5979	0.4978	0.6667	0.8594	0.7762	0.7419	0.8603
(c)	✓	✓		0.5737	0.4788	0.3430	0.5737	0.8002	0.7848	0.7071	0.8255
(d)	✓	✓	✓	**0.6745**	**0.6119**	**0.5191**	**0.6808**	**0.8840**	**0.8065**	**0.7714**	**0.8840**

Ablation studies: We conduct two series of ablation studies from the perspectives of loss functions and model components on the *Handwritten* and *Hdigit* datasets. Table 3 presents the ablation studies of different loss functions, where $\mathcal{L}_r$ is the loss to reconstruct original samples, $\mathcal{L}_m$ is the loss to capture key components of clusters, and $\mathcal{L}_f$ is the loss to fuse the extracted representations at the cluster-level based on the alignment results. Furthermore, Table 4 records the ablation studies of different model components, where C_h represents assigning a hyperedge group for each cluster, C_t represents the cluster-specific topology-guided cluster alignment, and C_i represents the designed cluster-aware operator in HGNNs. According to Table 3–4, we find that:

(1) In Table 3, (E) outperforms (A), indicating that autoencoders extract high-quality latent representations for subsequently extracting cluster-aware representations. Meanwhile, (E) outperforms (B), indicating that comprehensively capturing the intrinsic clustering structure of samples and infusing them into the feature extraction process effectively enhances cross-view alignment and fusion. Moreover, (E) is superior to (C), indicating that our cluster-level alignment and fusion strategy yields fused representations with a clearer cluster structure, thereby enhancing overall performance.

(2) In Table 4, (a) replaces the hyperedge group with a single hyperedge for each cluster, (b) replaces our designed topology-based alignment strategy with a cluster centroid-based strategy, and (c) removes the cluster-aware operator $\mathbf{\Psi}_c(\cdot)$. According to Table 4, (d) outperforms (a), indicating that our designed hyperedge group-based cluster modeling strategy effectively infuses the intrinsic clustering structure of samples into the HGNNs feature extraction process, which extracts cluster-aware representations that facilitate subsequent alignment and fusion. Meanwhile, (d) is superior to (b), indicating that our alignment strategy based on cluster-specific topology effectively performs cross-view cluster-level alignment for promoting subsequent fusion. Moreover, (d) is superior to (c), indicating that our designed cluster-aware interpolation operator facilitates the extraction of high-quality cluster-aware sample representations by the Hypergragh Neural Networks.

5 Conclusion

In this paper, we propose a hypergraph-based cluster-aware alignment clustering framework named HG-UMC for unaligned multi-view clustering. Different from existing unaligned multi-view clustering methods, HG-UMC constructs multiple groups of learnable hyperedges to capture the intrinsic clustering structure of samples precisely. These cluster-aware hyperedges are incorporated into the feature extraction process of the hypergraph neural network, enabling it to learn cluster-aware representations. Moreover, our proposed method performs cluster-level alignment based on cluster-specific topology, which significantly enhances the quality of alignment and fusion, thereby boosting the final clustering accuracy. Extensive comparative experiments with other state-of-the-art methods and further analysis on various benchmark datasets have verified the superiority of our proposed method.

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